

Parking Management System

Software Requirements Specification

SRS Version 1.0

Team T09

12 February 2026

Roxanne Krause (manager)

Eliud Garcia Reza

Ayman Hassen

Gabriel Ramirez

CS 460 Software Engineering

TABLE OF CONTENTS

Introduction	2
General Description	3
Specific Requirements	5
External Interfaces	5
Control Logic	9
Design Constraints	14
Definition of Terms	15
References	17

1 Introduction

This document outlines the Software Requirements Specification (SRS) for the Parking Management System (PMS), a multistory parking structure management solution. The purpose of this document is to specify the functional and non-functional requirements that will guide the system's implementation, ensuring it meets the objectives of parking efficiency, consistent and accurate tracking mechanisms, and administrative control. Most importantly, it will detail how the software will interact with the physical hardware to manage vehicle flow, parking states, and system constraints.

The implementation focus of PMS centers on utilizing a multitude of hardware components to achieve its established objectives. The system will rely on inputs from binary occupancy sensors, such as weight and ultrasonic sensors, installed at each parking spot to detect vehicle presence. At the entrance and exit, gate induction loop sensors and consequent vehicle detection will manage access. To communicate availability to drivers, the system will control multi-level LED displays at the structure entrance and on each floor, as well as individual spot indicator lights (i.e, RGB LEDs). These hardware elements from the sensory and gate network that the PMS software will monitor and control to maintain accurate capacity counts, provide inclusive visual cues for specialized spots (i.e, EV, handicap), and enable efficient installation and maintenance.

The document contains five sections describing the requirements specifications of the PMS. Section 2 provides a high-level overview of the entire system through a block diagram, illustrating the connections between input/output device drivers, the power supply, and the PMS software. Section 3 details the system's functional behavior, which is divided into detailing the external interfaces (listing all input events and output actions related to physical devices) and then featuring OMT dynamic diagrams to model system protocols, state changes, and the global lifecycle. Section 4 summarizes the key technical and operational limitations that shape the system's design and deployment. Finally, Section 5 details the term definitions that are used throughout this document for clarity.

2 General Description

The following figure, *Figure 1*, offers a block diagram of the connected physical systems that will interact through a network of input/output signals translated as necessary by device drivers. This offers a high-level visualization of the entire system, along with how all devices will be interacting with the PMS logic controller software.

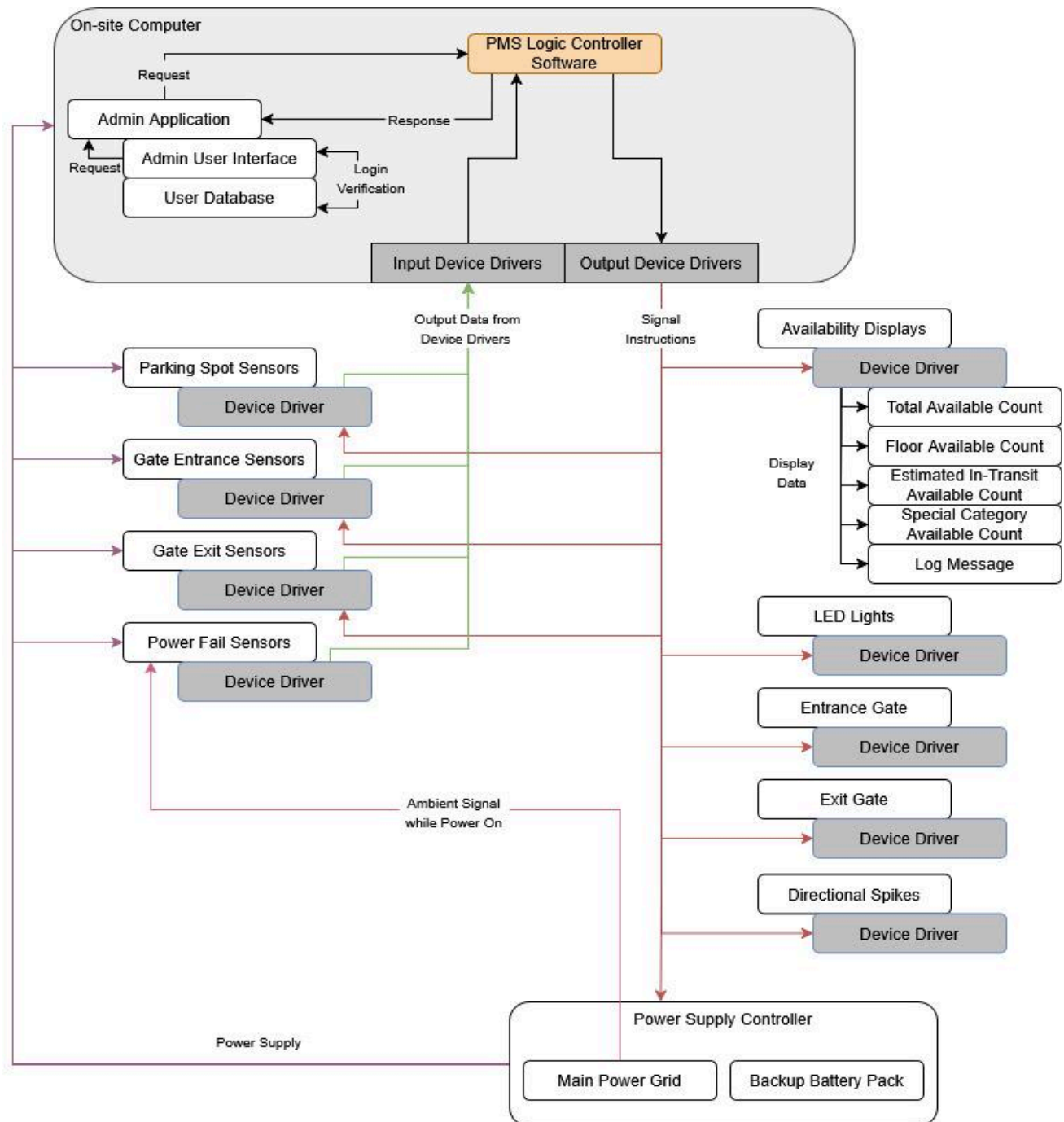


Figure 1: Overview of PMS Components and Devices

Although Section 3.1 will provide further detail on each device separately, *Figure 1* allows for a broader view of the system from a higher-level scope.

As shown, all interactions with the hardware devices require device drivers. The devices are largely defined as either input or output devices, with slight overlap in some of the input devices in specific conditions or commands, which will be shown in more detail in Section 3.2. However, the main input devices, or the sensors seen in *Figure 1*, will be able to relay obstruction signals through their device driver. The device driver will then transmit the signal to the PMS, received via an input device driver endpoint, to use for automated, software-driven tracking control. Likewise, the output devices—principally availability displays, gates, and LEDs—will receive signal commands from the PMS based upon automated logic and administrative override alike. These commands will be received by their device drivers and translated to the hardware accordingly.

The PMS itself and the administrative interface—to be executed as needed—will be running on an on-site machine. This will ensure that there is no wireless network communication required, which is a primary concern with any concrete structures (or other wireless-signal inhibiting materials)[2] used in the parking lot, along with other potential environmental impacts on reliable system communication. This will allow administrators, with database-verifiable login information, to have direct access to issue requests to the PMS without any outside oddities or interference. Further, the PMS will relay state information as changes occur or are requested to the administration application to provide a visual aid in the state of the lot, aiding the administrator in their tasks.

Finally, all systems will be included in a power system connected to the main power grid. A power sensor will be used to alert of any power failure. In this unique case, the PMS will signal a power controller to switch to a backup battery pack, keeping the system running through an emergency power outage.

3 Specific Requirements

The PMS has specific requirements regarding the system's functional behavior and the capabilities of the physical devices. This section describes the external interfaces that the PMS must be compatible with and details the rules of the control logic of the PMS through their respective use cases and given dynamic diagrams.

3.1 External Interfaces

The PMS depends upon external interfaces that provide it with the necessary feedback to issue commands and track the parking lot effectively. In the following sections, the flow of communication, as well as the specific interactions of each component shown in *Figure 1*, are more deeply expanded upon.

3.1.1 External Interface Communication

The external interfaces of the PMS establish the devices that allow the PMS software to know about and control the overall state of the system. In this section, the interfaces for the PMS are organized into input and output interfaces, where all of these interfaces facilitate digital communication. The PMS will process the information it receives and react when needed through the delegation of output action to certain external interfaces. *Figure 1* shows the major physical devices and baseline components that will be necessary to drive the PMS through a connected network. Meanwhile, *Figure 2* depicts at a lower level how this interaction will occur between the hardware devices and PMS.

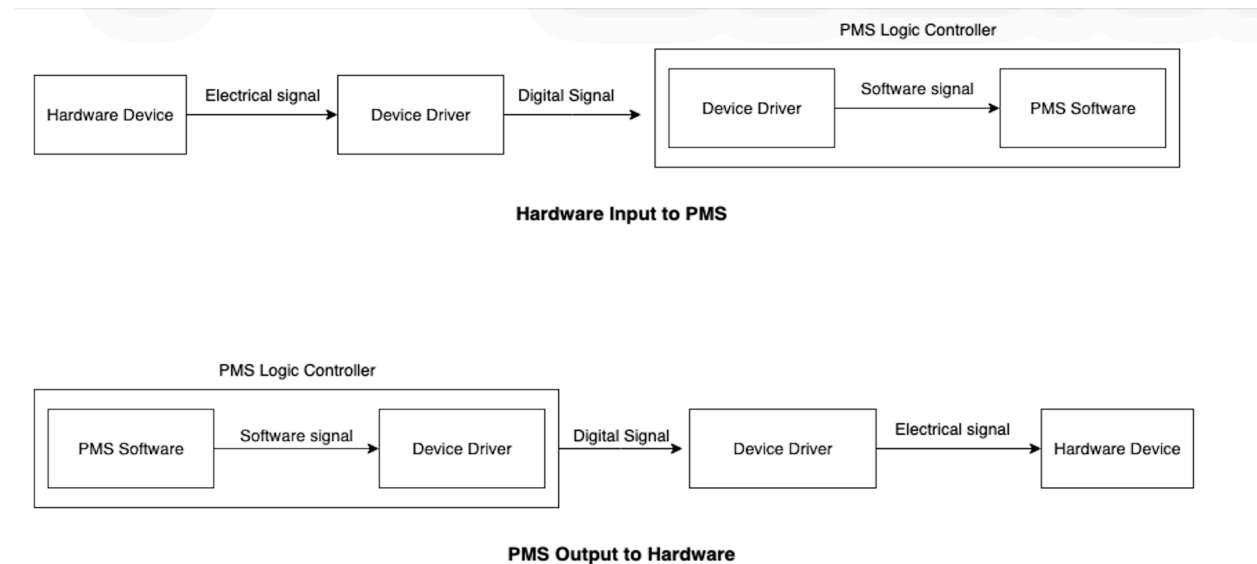


Figure 2: Input/Output Hardware Signal Sequence

As *Figure 2* shows, hardware devices are tightly connected to a device driver, which is used to translate an electrical signal to an understandable digital signal for the PMS to use within its software layer. The signal will be sent through a wired connection to a specific input driver, where the signal is then translated into a palatable software signal for the software portion of the system. PMS commands to the hardware have a similar flow in the other direction: translated software signals sent through an output device driver are received and transformed to low-level hardware electrical signals to carry out physical updates to the system.

The administrative application is, however, a special component of the system that is instead a software program built to run in tandem with the PMS software. *Figure 3* illustrates the connection that the administrative interface has with the PMS.

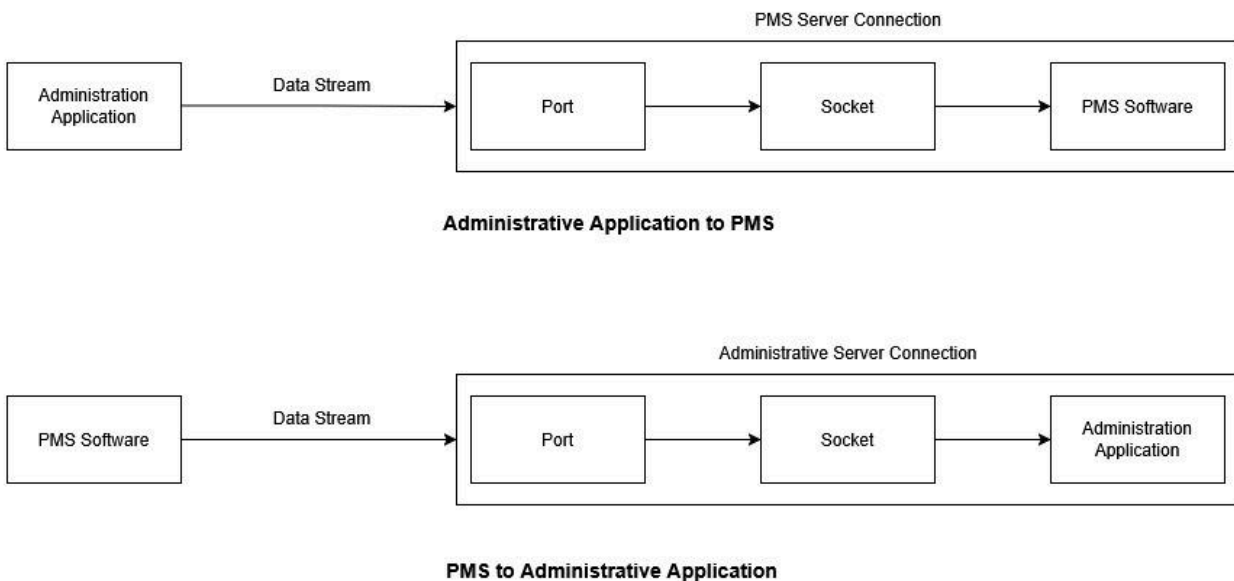


Figure 3: PMS/Administrative Application Intercommunication

Upon initialization of an instance of the administrative application, there will be a connection established upon a specific listening socket that the PMS software will consistently act as a server upon. The application will set up a server socket as well to receive automatic state-change notifications regularly from the PMS software in order to enable real-time updates to visual feedback and states of the application as it is running. In addition to the automatic update signals relayed by the PMS, the more intricate request/response architecture of communication between the two is elaborated further in Section 3.2.4.

Although this provides an overview of how major communication between the interfaces is needed for the PMS to succeed, the following sections will detail more specifically the interface specifications and what input the PMS will receive.

3.1.2 Sensor Inputs

Interface	Input to PMS
Parking Spot Sensors - a pair of weight and ultrasonic sensors positioned below and above, respectively, within each parking spot; continuously track the presence/absence of a vehicle within its associated parking spot.	When a vehicle is detected/undetected in the parking spot, this sensor signals PMS, indicating the current spot's availability status.
Gate Entrance Sensors - a pair of induction sensors positioned at the entrance gate of the structure; the sensor triggers after cars enter the parking structure - detects vehicles with the first sensor, then verifies the entry with the second one.	When a vehicle is detected/undetected at the gate, these sensors signal PMS, indicating a vehicle has requested entry to the structure and has successfully entered.
Gate Exit Sensors - a pair of induction sensors positioned at the exit gate of the structure; sensor triggers after cars exit the parking structure - detects vehicles with the first sensor, then verifies the exit with the second one.	When a vehicle is detected/undetected at the gate, these sensors signal PMS, indicating a vehicle has requested exit from the structure and has successfully exited.
Power Fail Sensors - continuously check for power availability within the overall structure to ensure the power is functioning.	Signals PMS of power disruptions once a prolonged loss of electricity is detected.

3.1.3 Administrative Application Inputs

Interface	Input to PMS
Admin User Interface - an interactive GUI allowing administrators to monitor and configure the different hardware related to PMS.	Relays allowed requests to view and modify PMS states. Refer to Section 3.2.4 and 4.3 for more detailed allowed requests.

3.1.4 Hardware Outputs

Interface	Output from PMS
Availability Displays - digital displays within and outside the structure; Displays the parking availability for each tracked type of parking spot and number of in-transit vehicles within the structure - at the floor-specific and structure level	Counts of availability per floor and structure, along with associated parking spot type, and an optional logging message from the administrator. Certain packages of graphical configurations modify icons related to specific information by its count.
LED lights - multi-colored RGB lights that can be switched depending on the specific parking spot type and availability/unavailability.	Signal containing a certain color to switch the LED.
Entrance Gate - a gate, with connected warning lights at the entrance that ceases to let vehicles in/out of the structure when closed; composed of a gate arm and lights for alerting drivers of movement nearby the directional spikes.	Signal to flash the gate lights for warning, and close/open the gate. An additional signal is optional to lock the entrance gate given an admin request.
Exit Gate - a gate, with connected warning lights, placed at the exit that ceases to let vehicles in/out of the structure when closed; composed of a gate arm and lights for alerting drivers of movement nearby the directional spikes.	Signal to flash the gate lights for warning and close/open the gate.

Directional Spikes - mechanical, but optionally digitally controlled, directional spikes that prevent vehicles from backing out of the entrance/exit gate areas. Additionally, it depends on the availability of structure.	Signals to raise or drop.
Parking Spot Sensors - The device driver is used to configure Parking Spot Sensors (see 3.1.1) - controls sensor power and configurations.	Signals to power off/on the sensor.
Gate Entrance Sensors - The device driver is used to configure Gate Entrance Sensors (see 3.1.1) - powers on/off the gate sensors	Signals to power off/on the sensor.
Gate Exit Sensors - The device driver is used to configure Gate Exit Sensors (see 3.1.1) - powers on/off the gate sensors	Signals to power off/on the sensor.

3.2 Control Logic

In this section, we present OMT design models to detail key dynamic processes of the PMS. The processes highlighted as follows primarily focus on the global parking lot state, gate control, parking spot tracking protocols, and administrative control.

3.2.1 Parking Lot Control Diagram

From the highest flow of control, the parking lot as a whole is modelled simply with two states. As the PMS runs, the parking lot state may be manipulated by the administrative application via a request to close. This will trigger actions carried out by the PMS to give an indication of closure while keeping the PMS running and continuing its essential tracking operations, allowing for a swift recovery once the lot is opened. *Figure 4* details the essential actions to allow for a simple lot state cycle as initiated by the admin.

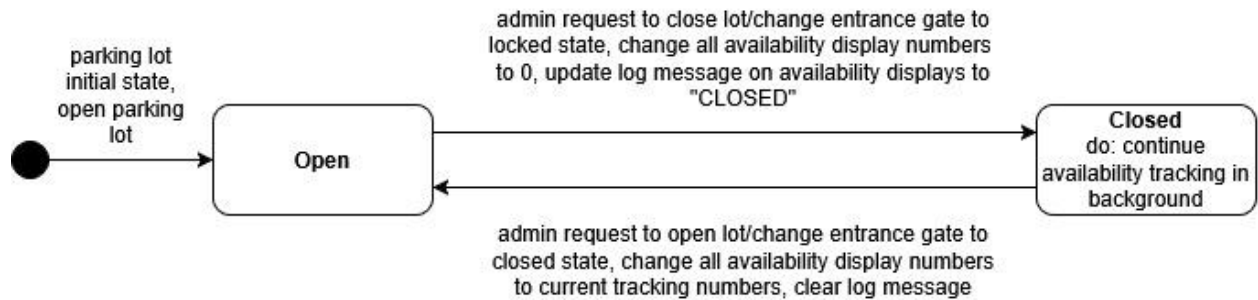
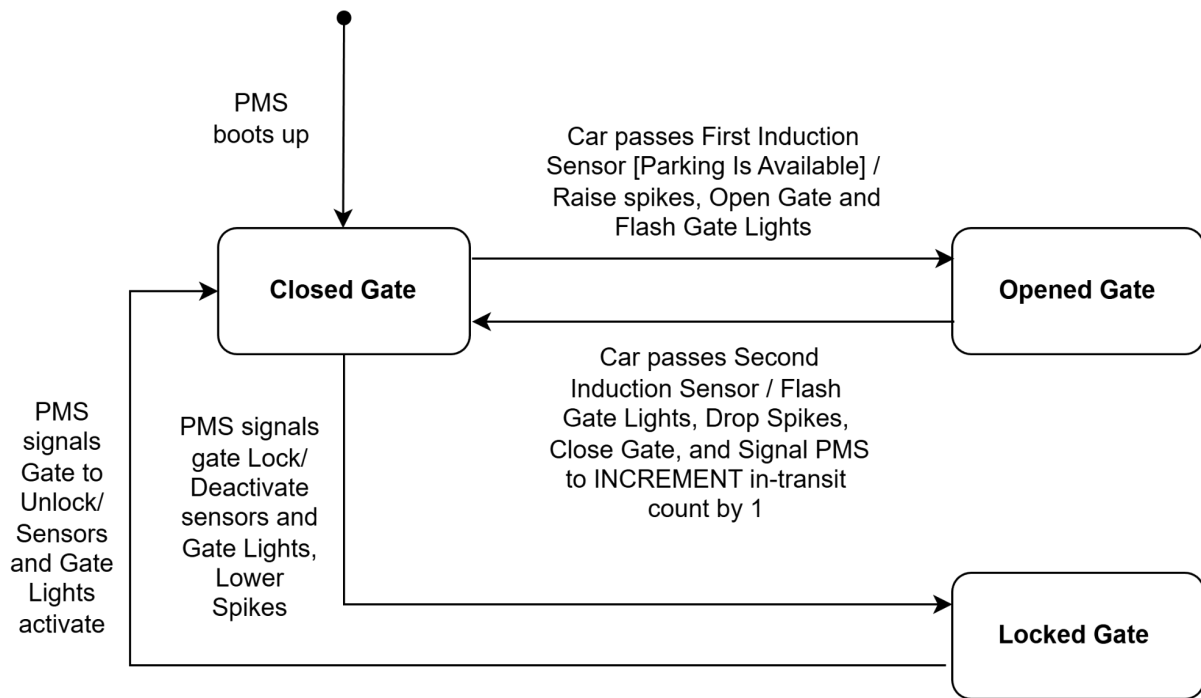


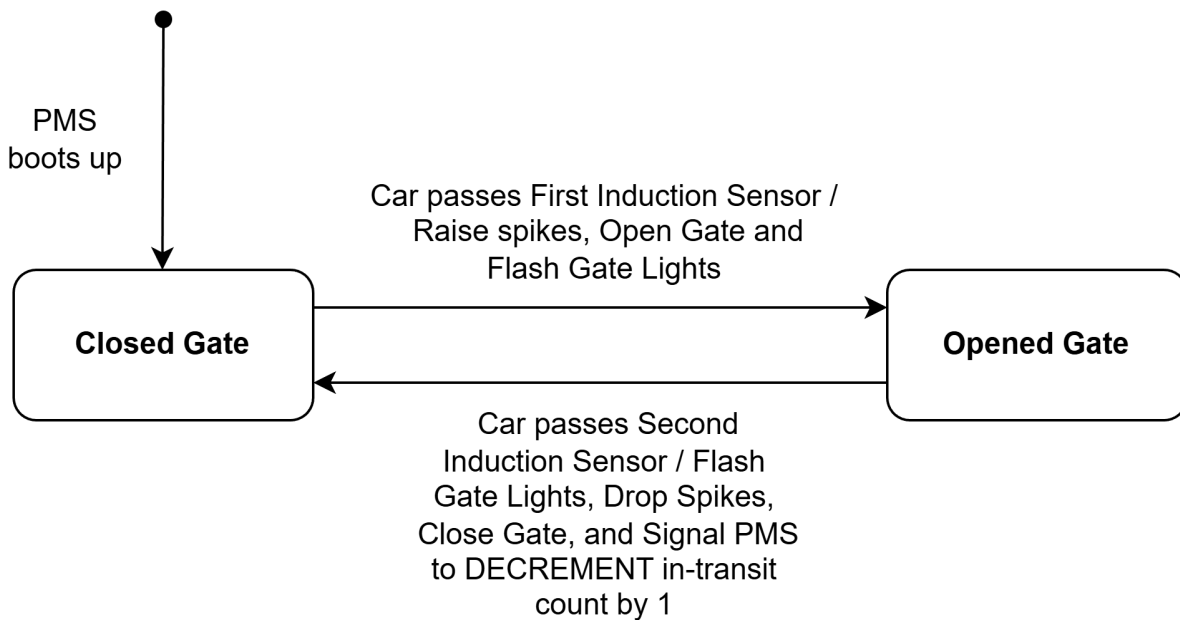
Figure 4: Parking Lot Dynamic Diagram

3.2.2 Gate Control Diagram

One of the major roles of the PMS is to assist in parking efficiency while maintaining reliability in tracking. In monitoring the entrance and exit gates, that objective is assisted. *Figure 5* focuses on the flow from entrance to exit of the driver, highlighting key events and conditions and the resulting actions taken by the PMS to maintain even traffic flow while maintaining up-to-date tracking.



(5.a) Entrance Gate Dynamic Diagram



(5.b) Exit Gate Dynamic Diagram

Figure 5: Entrance and Exit Gate Dynamic Diagrams

3.2.3 Parking Tracker Control Diagram

Likely the largest role for the PMS to play, the automated tracking of parking lot and spot states and availability is a complex process with high variability. The following figure, *Figure 6*, illustrates the most important driver action: parking. This diagram captures the key states and events through the process and what actions those will elicit from the PMS, along with dynamic administrative overrides as needed.

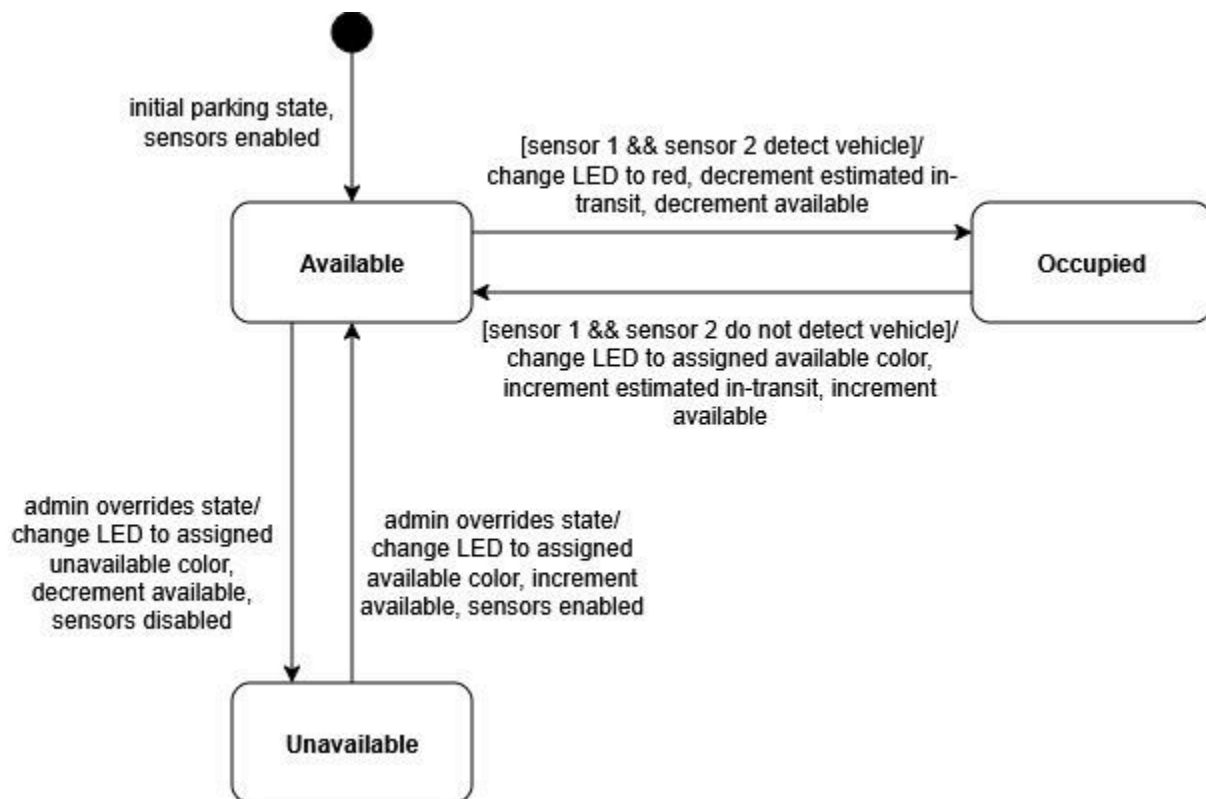


Figure 6: Parking Spot State/Availability Tracking Dynamic Diagram

3.2.4 Administrative Control Diagram

The administrative application is a key component for the PMS (although never required for baseline functionality). In order to provide administrative access for any necessary overrides, the application will be able to access special PMS functionality as needed through a request/response architecture. *Figure 7* will first show an overview of the lifecycle of the application and how it interacts with the PMS outside of expected receiving of state change notifications.

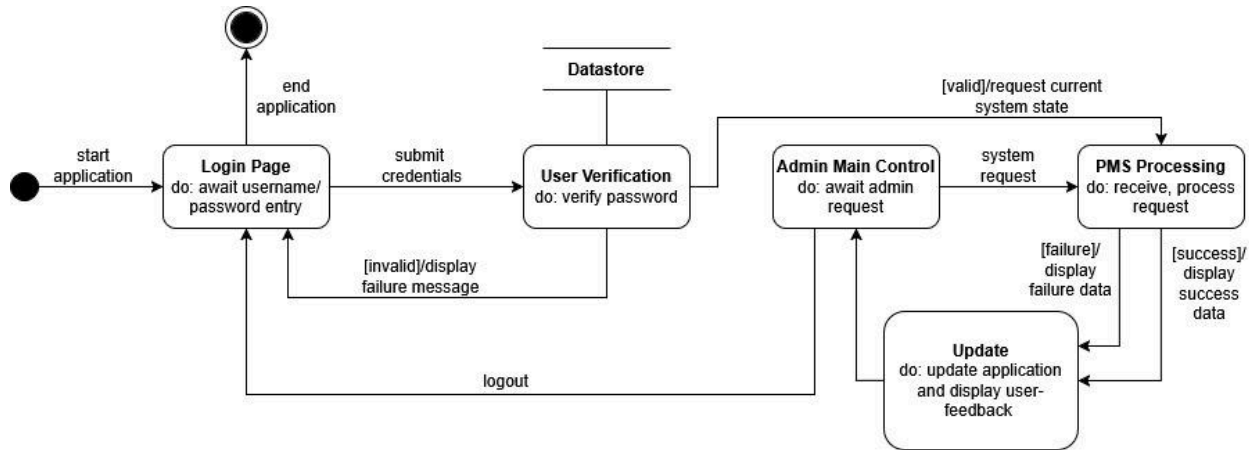


Figure 7: Overview of Administrative Application Lifecycle

For more details, *Figure 8* provides a diagram showing the dynamic actions that an administrator can take within the application.

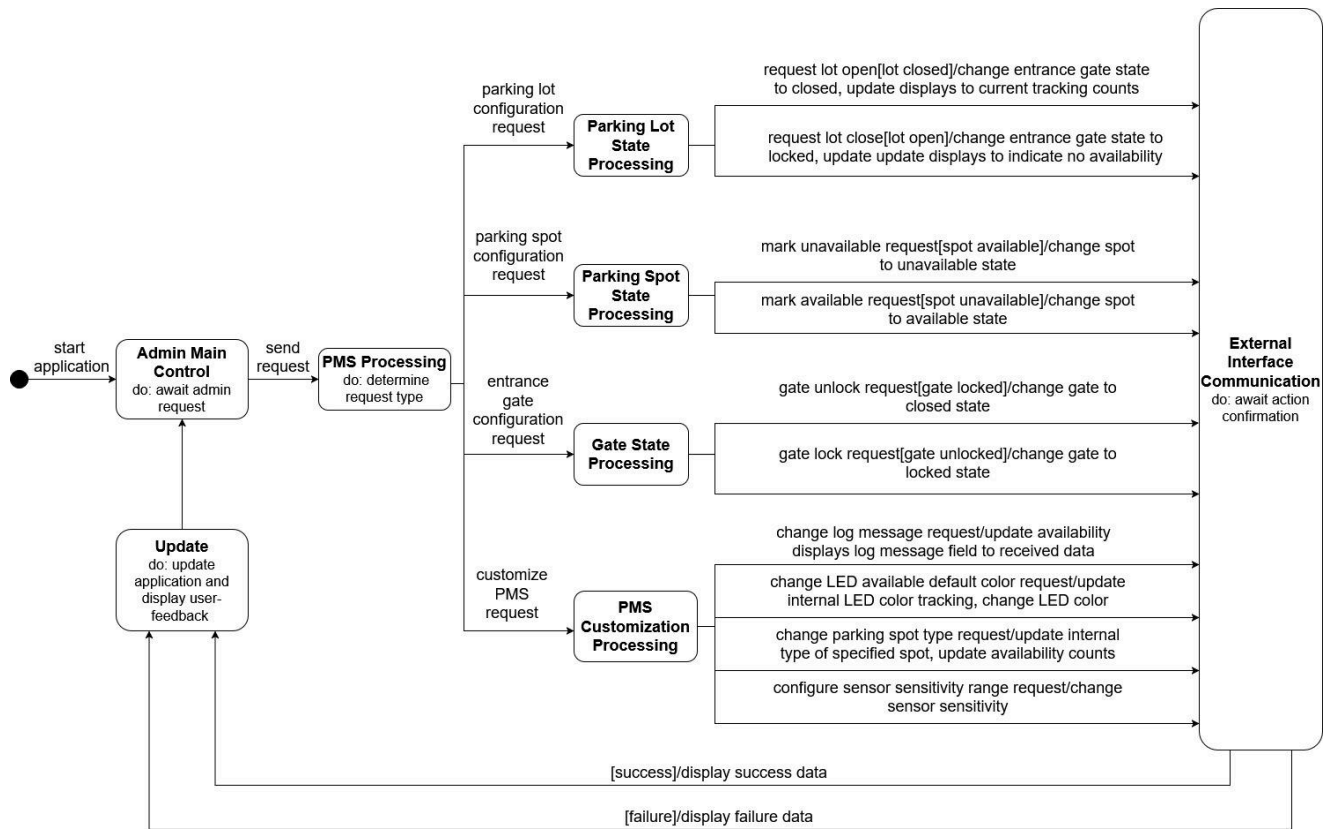


Figure 8: Administrative Requests

4 Design Constraints

The design constraints, as follows, are to be adhered to carefully in the development and consideration of the PMS. These constraints will support what is outlined in the Requirement Definition Document while adding specificity to the software and hardware constraints for the system.

4.1 PMS Hardware Constraints

- There must be a server on-site to run both the PMS software and the administrative application.
- All hardware intercommunication must be connected through a hard-wired network. There must be no reliance on communication of components via wireless network, such as Wi-Fi or Bluetooth, in order to compensate for known network-impeding building materials, such as concrete [2].
- Sensors are expected to have an accurate working range of at least -20°C to +50°C.
- Sensors must not be highly impacted by inclement weather (rain, fog, snow, etc.).
- All hardware must maintain a consistent state through a power outage to provide the reliability expected by the administration. Therefore, there must be a backup power source.
- The availability display must have a logging message output field available for custom administrative messages to be displayed, such that, the information remains bright and visible regardless of weather conditions..

4.2 PMS Software Constraints

- PMS software must be able to receive requests and send responses with the administrative application (if it is running). This communication must reside through designated server ports/sockets.
- All hardware must have an associated device driver to receive digital commands from the PMS.
- Software must all be written using the programming language C++, using it for full logical and user-interface development.

4.3 Administrative Application Constraints

- Actions that the administrator must be able to carry out manually through the administrative application are:
 - Changing the status and type of a parking spot.
 - Changing the status of the parking lot.
 - Configuring the LED light to a needed color or state.
 - Altering the status of the entrance gate by locking it effectively, restricting traffic into the facility as needed.

- Configuring the settings of sensors by altering the scope of sensitivity based on vehicle type/unique need.
- Altering the custom message of the availability displays.
- The user interface of the administration application must be clear and easy to use. Further, a graphical display of each floor must be easily accessible and clear for oversight needs.
- The user interface must update dynamically as a result of state changes communicated by the PMS.
- There must be a username and password verification to access the security application.
- There must be a secure database used for password verification for the application.
- There must be an administrator on-site to use the application. There is no remote access permitted.

5 Definition of Terms

In this section, we define some relevant terms that are used frequently throughout the paper for accurate communication. The reader can use this section as a future reference while going through the document.

Administrator - Generally, owners of the structure or specifically assigned personnel in charge of managing the PMS, enabling manual override of certain functionality, and manually updating parking space availability whenever necessary.

Administrative Application - The software application that is provided for administrators to make manual changes to the PMS. Sometimes simply referred to as an *application*.

Availability - The status of a parking spot, which will be either filled (vehicle parked) or available (no vehicle parked).

Availability Display - The required main display is made available in front of the parking lot to provide availability counts for various spot types. Additionally can refer to displays on each floor of the lot displaying the same information as the aforementioned main display.

Bluetooth - A short-range, wireless technology standard that enables different devices to connect over radio frequencies without the use of cables.

C++ - A high-performance programming language that is an extension of the C programming language by adding object-oriented functional features used to build complex software.

Device Driver - A software translation layer that allows for an interface between digital signals received and instructions for the hardware it controls [1].

Driver - Individuals who drive a vehicle (of any type) and desire to use a parking space for their vehicle.

EV - Electric Vehicle. Requires a designated charging station to recharge.

GUI - “Graphical User Interface”. Primarily refers to the front-end application that the administrator will issue requests to.

Handicapped - Parking space for individuals who have a condition that restricts ease of movement. It can also be referred to as an accessibility space.

In-Transit - A status of a vehicle referring to when the vehicle is within the garage, but not parked. Additionally, the count of these is the number of vehicles with this status.

LEDs - A light-emitting diode, used for indicating parking space availability.

OMT - “Object Modelling Technique”. A specific format for modelling dynamic processes of a system.

Port - A virtual endpoint within an operating system that acts as a specific communication doorway that can allow network traffic to be directed to the right destination.

PMS - The Parking Management System, consisting of all the devices and software necessary to manage and oversee parking lot operations.

RGB - A color system that uses red, blue, and green to blend into a wide array of other colors.

Server - A computer software system that can manage network resources and deliver functionality to other devices over a network.

Spikes (Spiked Speed Bump) - A controlled traffic-calming device used to direct traffic to one direction using spikes and a raised bump. Occasionally referred to as *directional spikes*.

Vehicle - a metallic, wheel-based machine primarily used for transporting people, including cars, trucks, motorcycles, etc.

Wi-Fi - A wireless networking technology that utilizes radio waves to provide high-speed internet without the need for physical cables.

6 References

[1] “Device Driver”, wikipedia.org, https://en.wikipedia.org/wiki/Device_driver.

Accessed Feb. 9th 2026

[2] “Electromagnetic Signal Attenuation in Construction Materials”, nist.gov,

<https://www.nist.gov/publications/electromagnetic-signal-attenuation-construction-materials>.

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